

ROLL NO.

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Subject Code & Name: CS3403 & Object Oriented Programming and System with Java
Program/Branch/Year: B.TECH 2nd Year/ SEM IV
Time: 2Hrs

Max Marks: 60

General Instructions. Read all instructions carefully.

1. Do not write anything on the question paper except your Roll No.
 2. Answers should be written in clear and legible handwriting. Neatly labeled diagrams will fetch better marks.
 3. Students must clearly write the question number & sub-part of question that they are attempting.
 4. This question paper consists of 3 questions and all questions are compulsory. All questions carry 20 marks each.
 5. Students are expected to take about 35 minutes each for Questions 1 to 3 and Remaining 15 minutes are for reading the question paper and revision
- Note: BL refers to Blooms Taxonomy Level indicating Level of Complexity of a question

Parts i) and ii) of each question are compulsory and each part carries 2 marks. Parts iii), iv) and v) carry 8 marks each and the student may attempt any 2 parts.

Q1.

- i) `int x=0,y=0,z=0;`
`x=(++x + y--)*z++;` what will be the value of x
- a) -2
 - b) -1
 - c) 0
 - d) 1

[2 marks,BL-2].

- ii) In java Byte, Short, Int and Long all of these are

[2 marks,BL-1]

- a) Signed
- b) Unsigned
- c) Both a and b
- d) None of above

- iii) Define an Array. Classify the types of arrays in Java. Explain how arrays are created and declared in java? Give an example.

[8 marks, BL-3]

- iv) Explain the loop control in java. Also write a Java program to calculate the factorial of a number. [8 marks,BL-2]

- v) Describe object-oriented methodology? Explain the features of OOPs in brief. [8 marks,BL-2]

Q2.

- i) If a function in java is declared private then it _____

[2 marks BL-2].

- a) Can't access standard output.
- b) Can access standard output.
- c) Can't access any output stream.
- d) Can access only the output streams.

- ii) _____ method cannot be overridden.
a) public
b) protected
c) final
d) none

[2 marks, BL-1]

- iii) Explain Constructor and its type. Also write a java program to implement constructor overloading in java. [8 marks, BL-2]
iv) Explain the Inheritance and its type. Why multiple inheritance not supported in java? Also write a program of inheritance. [8 marks, BL-2]
v) Differentiate between class, abstract class, and interface. Justify your answer with an example. [8 marks, BL-4]

Q3.

- i) Find out the exception which is not type of the exception. [2 marks, BL-2]
a) Checked Exception
b) Unchecked Exception
c) Runtime Exception
d) Static Exception

ii) Find out the method which is used to pause the execution of a thread for a specific amount of time. [2 marks BL-1]

- a) sleep()
b) pause()
c) wait()
d) yield()

iii) Explain the exception handling and use of try catch block with an example program. [8 marks, BL-2]

iv) Explain the multithread, life cycle of thread with example. [8 marks, BL-2]

v) Discuss various access protections in packages, with example. Also explain the procedure to create package in java. [8 marks, BL-3]

MID SEMESTER EXAMINATION

ROLL NO.

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Subject Code & Name: CS 3404 Theory of Automata & Formal Language

Program/Branch/Year: B.TECH 2nd Year/ SEM III

Time: 2 Hrs

Max Marks: 60

General Instructions. Read all instructions carefully.

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Note: BL refers to Blooms Taxonomy Level indicating Level of Complexity of a question

Parts i) and ii) of each question are compulsory and each part carries 2 marks. Parts iii), iv) and v) carry 8 marks each and the student may attempt any 2 parts.

Q1.

i) A Language for which a DFA exist is a _____

[2 marks,BL-1].

- a) Regular Language
- b) Non-Regular Language
- c) May be Regular
- d) Cannot be said

ii) A regular language over an alphabet Σ is one that cannot be obtained from the basic languages using the operation

[2 marks, BL-1].

- a) Union
- b) Concatenation
- c) Kleene*
- d) All of the mention

iii) Define DFA, NFA, String, Language and Grammar with suitable explanation

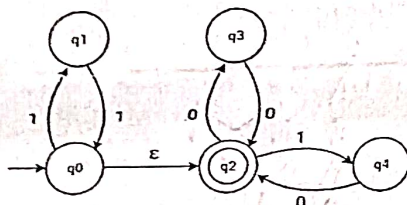
(8marks, BL-3)

iv Draw a DFA to accept string of 0's and 1's ending with the string 011.

[8 marks BL-2]

v) Convert the following epsilon NFA to NFA .

(8 marks,BL3)



Q2.

i) Given: $\Sigma = \{a, b\}$ $L = \{x \in \Sigma^* \mid x \text{ is a string combination}\}$ Σ^4 represents which among the following? [2 marks BL-2].

- a) $\{aa, ab, ba, bb\}$
- b) $\{aaaa, abab, \epsilon, abaa, aabb\}$
- c) $\{aaa, aab, aba, bbb\}$
- d) All of the mentioned

[2 marks BL-2].

ii) All the regular languages can have one or more of the following descriptions:

i) DFA ii) NFA iii) e-NFA iv) Regular Expressions

Which of the following are correct?

- a) i, ii, iv
- b) i, ii, iii
- c) i, iv
- d) i, ii, iii, iv

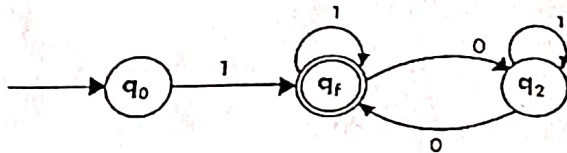
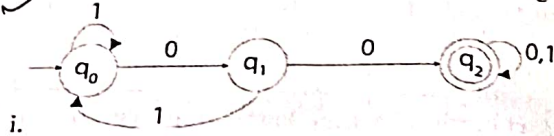
iii) Write formal definition of regular expression with suitable example. State Arden's theorem and its use.

(8 marks, BL-3)

iv) State the pumping lemma theorem for regular sets. Prove that $L = \{a^i b^i \mid i \geq 0\}$ is not regular.

(8 marks, BL3)

v) Create the regular expressions for the state diagram given below.



(8 marks, BL-4)

Q3.

i) A language is accepted by a push down automata if it is:

[2 marks BL-2].

- a) regular
- b) context free
- c) both a and b
- d) none of the mentioned

ii) A push down automaton employs _____ data structure

[2 marks BL-1].

- a) Queue b) Linked List c) Hash Table d) Stack

iii Explain push down automata with an example. *Difference b/w CFG*

(8 marks, BL 2)

iv) for the given CFG give leftmost and rightmost derivations of the following strings a) 00101 b) 10011 c) 00011

(8 marks, BL-3)

$S \rightarrow A1B, A \rightarrow 0A \mid \epsilon, B \rightarrow 0B \mid 1B \mid \epsilon$

v) Construct PDA: For the language $L = \{a^n b^{2n} \mid a, b \in \Sigma, n \geq 0\}$

(8 marks, BL-3)

neha
22/3/24

Chiranjit
22/3/24

Ding
22/3/24

Adite Kharghar
28/03/24

QUANTUM UNIVERSITY

MID SEMESTER EXAMINATION

Session 2023-24 (Odd Semester) October 2023



ROLL NO.

2	2	0	1	3	0	1	0	6	4
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Subject Code & Name: CS3402 & Computer Network
Program/Branch/Year: B.Tech 2nd CSE / SEM 3
Time: 2Hrs

Max Marks: 60

General Instructions. Read all instructions carefully.

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Q1.

i) The first Network was called _____

- a) CNNET
- b) NSFNET
- c) ASAPNET
- d) ARPANET

ii) The required resources for communication between end systems are reserved for the duration of the session between end systems in _____ method. (2 marks)

iii) Explain guided media differ from unguided media. Explain the three types of guided media and two types of unguided media. (8 marks)

iv) Differentiate circuit switching and packet switching?

v) Define different types of transmission mode and also its types.

(8 marks)

(8 marks)

Q2.

i) is a type of network interface.

ii) the layer of the OSI model is responsible for error-free data transmission

(2 marks)

(2 marks)

(8 marks)

iii) Explain Data Link layer, and its primary function in the OSI model?

iv) Describe the difference between a MAC address and an IP address. How are they used in the Data Link layer and the Network layer, respectively?

(8 marks)

v) Explain the term flow control, and how is it implemented in the Data Link layer? What are the advantages and disadvantages of different flow control methods? (8 marks)

Q3.

i) layer of the OSI model is responsible for routing data between different networks. (2 marks)

ii) Name the addressing scheme which is used by the IP protocol? (2 marks)

a) MAC addresses

b) IP addresses

c) URL addresses

d) DNS addresses

iii) Explain the role of the network layer in the OSI model? (8 marks)

iv) Name three network layer protocols used for routing data between networks and also explain in detail. (8 marks)

v) Compare and contrast IPv4 and IPv6, including their addressing schemes and any limitations or benefits. (8 marks)

12/10/23

Shrey
Aishwarya

QUANTUM UNIVERSITY

MID SEMESTER EXAMINATION

Even Semester April 2023

ROLL NO.

2201301064

Subject Code & Name: CS3401 Software Engineering
Program/Branch/Year: B.Tech 2nd Year/ SEM IV
Time: 2 Hrs

Max Marks: 60

General Instructions. Read all instructions carefully.

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Q1.

- i) After the design phase the document prepared is known as _____ [BL-1](2 marks)
- ii) CASE stands for _____ [BL-1] (2 marks)
- iii) State and discuss the various myths related to the software. [BL-2](8 marks)
- iv) Explain the various phases of SDLC model in details. [BL-2] (8 marks)
- v) Compare and contrast the waterfall and agile development methodologies with diagram. [BL-2] (8 marks)

Q2.

- i) In waterfall model, output of one phase is input to next phase (True/False) [BL-1](2 marks)
- ii) Selection of life cycle model is based on _____ [BL-1](2 marks)
- iii) Explain the primary purpose of a (SRS) in detail with example. [BL-2] (8 marks)
- iv) Differentiate between the user and system requirements in detail. [BL-4] (8 marks)
- v) Explain working of SEI-CMM model in detail with diagram. [BL-2] (8 marks)

Q3.

- i) In size oriented metrics, metrics are developed based on the _____ [BL-1](2 marks)
- ii) The following is not a direct measure of SE process [BL-1](2 marks)
 - a) Efficiency
 - b) Cost
 - c) Effort Applied
 - d) All of the mentioned
- iii) Explain the characteristics of a good design and also describe different types of coupling and cohesion. [BL-1](2 marks)
- iv) Explain the Iterative Model in detail with proper diagram. [BL-2] (8 marks)
- v) Write short note on: [BL-2] (8 marks)
 - a) Design documents
 - b) Software crisis.
 - c) Spiral model

[BL-3] (2 + 2 + 4 marks)

[Signature]